

Pabel Shahrear
Department of Mathematics
Shahjalal University of Science and Technology
Sylhet, 3114, Bangladesh

September 4, 2026

Editor-in-Chief
Journal of Computer Languages
Elsevier

Dear Editor-in-Chief,

We are pleased to submit our manuscript entitled **“Conformance-Gated Code Generation: An Empirical Benchmark of Architectural Specification Across LLM Families”** for consideration as a Research Article in the *Journal of Computer Languages*.

Our research systematically evaluates how structured architectural specification artifacts influence large language model (LLM) performance, specifically measuring architectural conformance, output token efficiency, and generation consistency across model scales.

Through a controlled study of 2,451 evaluations spanning 30 backend feature prompts, two generation conditions, and five model families ranging from 7B to over 100B parameters, we evaluate the Aether Framework—a conformance-gated system that injects a machine-readable architectural specification into the LLM prompt. The central finding is a capacity-dependent “funnel effect” with an apparent threshold between 8B and 12B parameters, providing empirical evidence on how the effectiveness of dense instruction artifacts depends intrinsically on model scale.

We confirm that this manuscript is original work, has not been published previously, and is not currently under consideration for publication elsewhere. All authors have read and approved the final version of the manuscript. We have no conflicts of interest to disclose.

Thank you for your time and consideration. We look forward to your response.

Sincerely,

Pabel Shahrear
Corresponding Author
Department of Mathematics
Shahjalal University of Science and
Technology
shahrear-mat@sust.edu